WASP-12b

R-1457

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-12_161102` · created 2026-08-01T10:19:04+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457694.83188 +/- 0.00033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.003 +/- 0.045
Transit depth (Rp/Rs)^2	0.00073 +/- 0.024 [%]
Orbital Inclination (inc)	81.83 +/- 2.91
Airmass coefficient 1 (a1)	1.26 +/- 0.59
Airmass coefficient 2 (a2)	-0.019 +/- 0.073
Scatter in the residuals of the lightcurve fit is	49.66 %
Best Comparison Star	#3 - [202, 243]
Optimal Aperture	3.0
Optimal Annulus	5.5
Transit Duration (day)	0.1088 +/- 0.0098

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.003 ± 0.045 vs 0.1178 (deviation 1.69σ)
Duration fit vs published	0.1088 d vs 0.125 d (deviation 1.09σ)
Mid-time O–C	0.5 min
Depth SNR	0.0
Residual scatter	49.66 %

Light curve

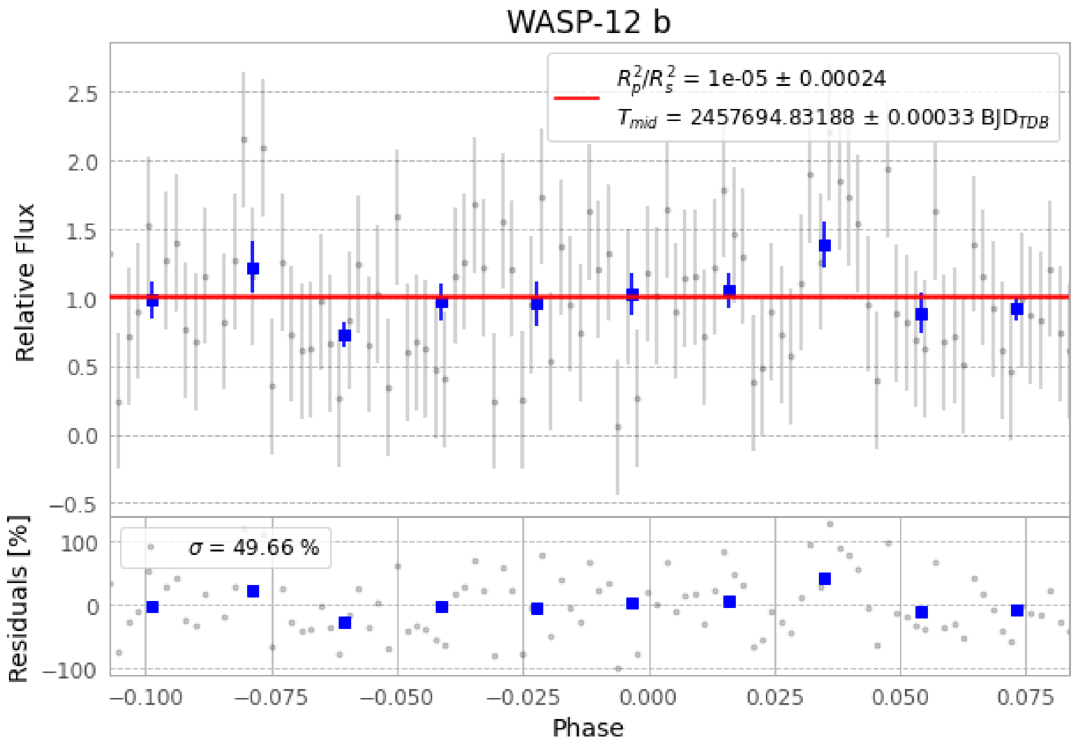

Figure 1 — FinalLightCurve_WASP-12 b_2016-11-01.png (SHA-256 dad390ab235fff05…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [319, 399], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 e390ffdebba6ec6f…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

101 FITS frames (66 MB), `WASP-12161102050312.FITS` → `WASP-12161102100312.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-161102.log (2ab52533fe8b…), FinalLightCurve_WASP-12 b_2016-11-01.png (dad390ab235f…), FinalLightCurve_WASP-12 b_2016-11-01.csv (30feecccd2cd…)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)